

Evaluation

Create an eval to measure date extraction

1. Manually extract due dates from 10-20 invoices

test invoice 1

per example ground truth



"August 20, 2025" → "2025/08/20"

2. Specify output format of data in prompt

*Format the due date as
YYYY/MM/DD*

3. Extract date from the LLM response using code

```
date_pattern = r'\d{4}/\d{2}/\d{2}'  
extracted_date = re.findall(date_pattern, llm_response)
```

4. Compare LLM result to ground truth

```
if (extracted_date == actual_date):  
    num_correct += 1
```

Driving your development process with evals

- Build a system and look at outputs to discover where it is behaving in an unsatisfactory way
 - E.g. incorrect due dates in invoice data extract
- Drive improvement by putting in place a small eval with ~20 examples to help you track progress
- Monitor as you make changes to workflow (e.g. new prompts, new algorithms) and see if the metric improves

Create an eval to measure text length

1. Create a set of 10-20 test tasks

Image



Example prompt

Create an Instagram post

Create an Instagram post

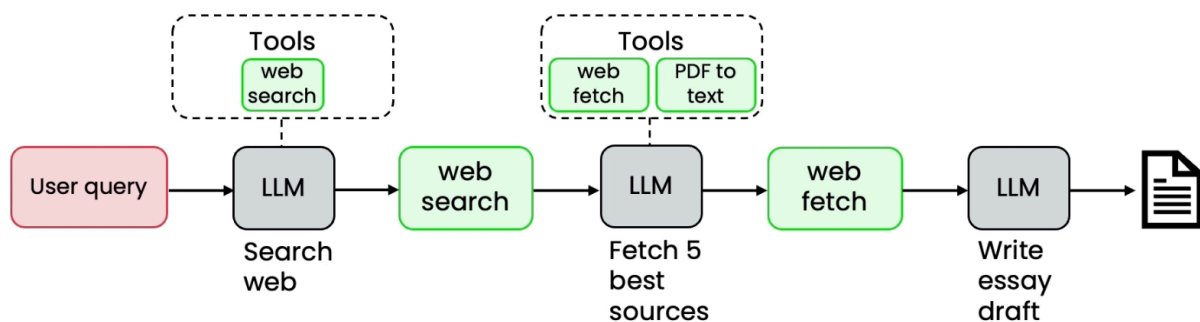
2. Add code to measure word count of the output

```
word_count = len(text.split())
```

3. Compare length of generated text to limit

```
if (word_count <= 10):  
    num_correct +=1
```

Example: research agent



Prompt	Issues
Recent black hole science	Missed high-profile result that had lots of news coverage
Renting vs buying a home in Seattle?	Seems to do a good job
Robotics for harvesting fruit	Didn't mention leading equipment company

Sometimes it misses the important info that humans would not have missed.

Create an eval to measure performance

- 1. Choose 3-5 gold standard discussion points for each topic
- 2. Use LLM-as-a-judge to count how many topics were mentioned
- 3. Get score for each prompt in eval set

Example prompt	Gold-standard talking points
Black holes	Event horizon, radio telescope
Robotic harvesting	RoboPick, pinchers

ground truth annotations

Determine how many of the 5 gold-standard talking points are present in the provided essay.

Original Prompt
{original_prompt}
Essay to Evaluate
{essay_text}
Gold Standard Talking Points
{gold_standard_points}
Output Format
Return a json object with two keys: score (a single number between 0 and 5), and explanation (a string that lists the talking points present)

Note: Gold standard point in the context of AI refers to the highest quality, most reliable, and validated reference point for training, evaluating, or implementing an AI system.

Two “axes” of evaluation

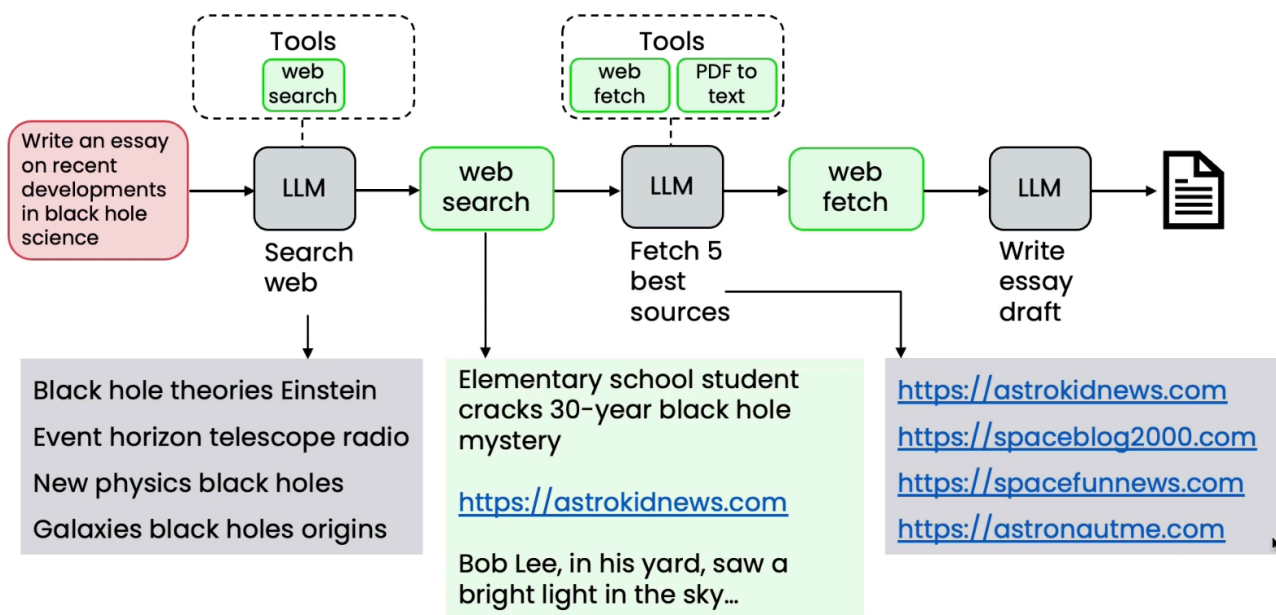
	Evaluate with code (objective)	LLM-as-judge (subjective)
Per example ground truth	Checking invoice date extraction <pre>if (extracted_date == actual_date): num_correct +=1</pre>	Counting gold-standard talking points Count the number of gold standard points in the following text...
No per example ground truth	Checking marketing copy length <pre>if len(text) <= 10: num_correct += 1</pre>	Grading charts with a rubric Grade this chart according to (i) whether it has clear axes labels, (ii)

Tips:

- Quick and dirty is ok to start
- As you find places where your evals fail to capture human judgement as to which system is better, use that as an opportunity to improve the metric.
- Look for a place where the performance is worse than human

Inspecting traces

Looking at traces



Counting up the errors

Prompt	Search terms	Search results	Picking 5 best sources	...	...
Recent developments in black hole science		Too many blog posts, not enough papers			
Renting vs buying a home in Seattle			Missed well-known blog		
Robotics for harvesting fruit	Terms too generic	Website for elementary school students			
...	...	...	...		
Batteries for electric vehicles		Only selected US-based companies	Missed magazine		
	5%	45%	10%	...	...

Tips for error analysis

- Develop a habit of looking at traces
- Carry out error analysis to figure out which component performed poorly, leading to a poor final output
- Use the error analysis output to decide where to focus efforts

Counting up the errors

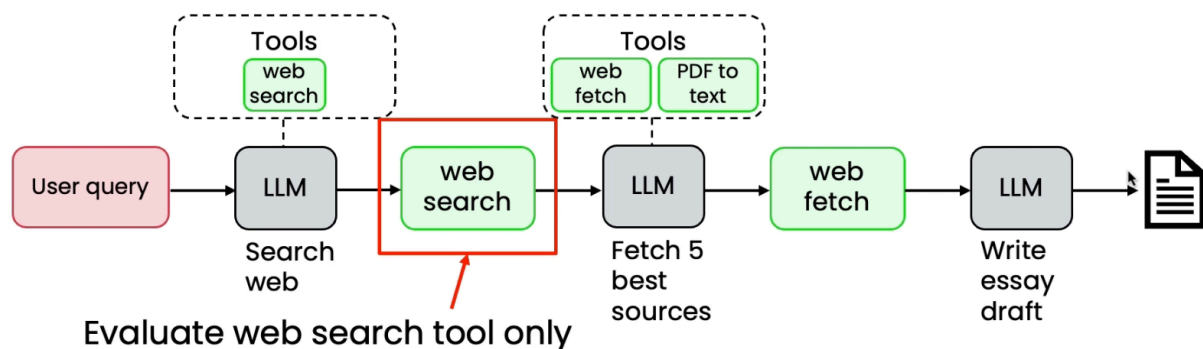
Input	LLM-drafted query	Orders database query	LLM-drafted email
Email 1	Wrong table		
Email 2		Error in database entry	Didn't address details of order
Email 3	Incorrect math		
	...	...	...
Email 50			Defensive tone

75% **4%** **30%**

So the drafted query is where we need to put effort in.

Component Evals

Example: research agent



- Create a list of gold standard web resources
- Write code that calculates how many results correspond to gold standard websites e.g. F1-score
- Track as you vary hyperparameters: e.g., search engine, number of results, dates

Benefit of Component-level evaluations

- Can provide a clearer signal for specific errors
 - avoid the noise in the end-to-end system
- More effective for a focused team to optimize
 - work on smaller, more targeted problems faster

Improving non-LLM component performance

ex: web search, text-retrieval for rag, code execution, trained ML model

- **Tune hyperparameters of component**

Web search: Number of results, date range
RAG: Change similarity threshold, chunk size
ML models: Detection threshold

- **Replace the component**

Try a different web search engine, RAG provider, etc.

- **Improve the prompts:** few-shot-prompting
- **Try new model:** try multiple LLMs and use evals to pick the best
- **Split up the step:** decompose the task into smaller steps
- **Fine-tune a model:** fine-tune on your internal data to improve performance

Instruction following

Summary of customer call:

On July 14, 2023, Jessica Alvarez (SSN: 555-44-3333) of 1024 Maple Ridge Lane, Boulder, CO 80301, submitted a support ticket....

Prompt

Identify all cases of personally identifiable information (PII) in the text below.

Then return a list of the identified PII classified by type, and then redact all the identified PII with "*****".

Separate the list and the redacted text with "REDACTED: ".
{text}

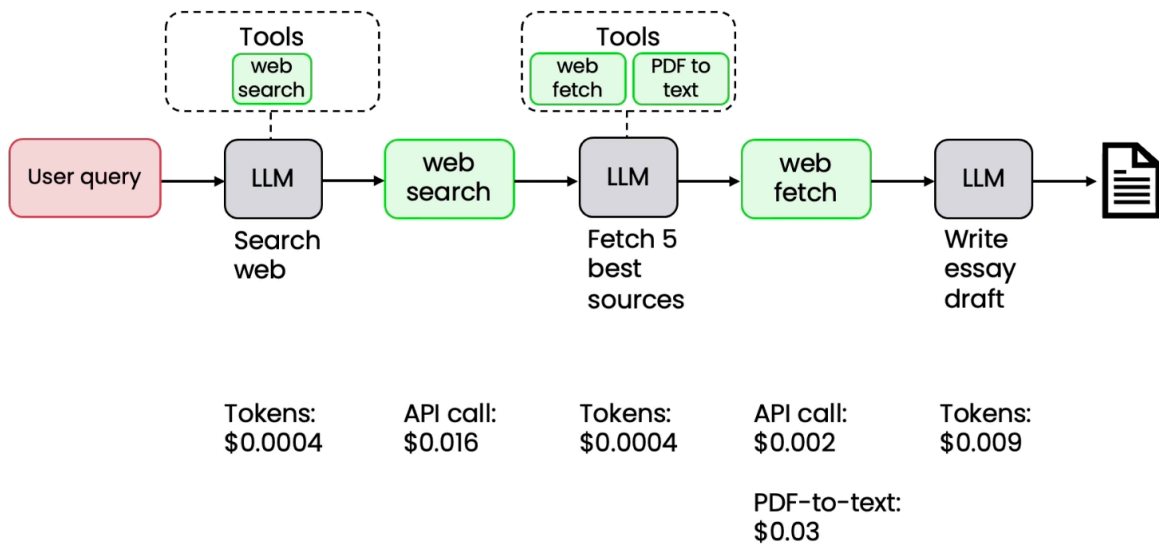
Intuition for model intelligence

- Play with models
 - Having a personal set of evals might be helpful
 - Read the other prompt for ideas
- Use different models in your agentic workflow
 - Which model work for which type of tasks
 - aisuite makes it easier to quickly swap out models

Cost for the workflow

- LLM steps (pay per token)
- Any API-calling tools (pay per api call)
- Compute steps (based on server capacity/cost)

Costing your workflow



Development process

- **Build**
 - build end-to-end system
 - improve the individual component
- **Analysis**
 - examine output; trace
 - build evals; compute metrics
 - error analysis
 - Component-level evals